

API TECHNICAL SPECIFICATION

DataStream Analytics Platform - REST API v2.4

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1. OVERVIEW

The DataStream Analytics REST API provides programmatic access to real-time data processing, analytics computation, and visualization services. This document covers authentication, endpoints, request/response formats, rate limiting, and error handling.

Base URL: <https://api.datastream-analytics.com/v2>

Content-Type: application/json

Authentication: Bearer token (OAuth 2.0) or API key (X-API-Key header)

2. AUTHENTICATION

2.1 OAuth 2.0 Flow (Recommended)

POST /oauth/token

Request Body:

```
grant_type: "client_credentials"
client_id: "your_client_id"
client_secret: "your_client_secret"
scope: "read write analytics:admin"
```

Response (200 OK):

```
access_token: "eyJhbGciOi..."
token_type: "bearer"
expires_in: 3600
scope: "read write analytics:admin"
```

2.2 API Key Authentication

Include in request header:

X-API-Key: dsa_live_k8j2m9x4p1q7...

API keys can be generated from the Developer Portal. Keys are scoped to specific permissions and can be rotated without downtime.

3. CORE ENDPOINTS

3.1 Data Ingestion

POST /v2/streams/{stream_id}/events

Description: Ingest events into a data stream

Rate Limit: 10,000 requests/minute

Max Payload: 1MB per request, 100 events per batch

Request Body:

```
stream_id: "user-activity-prod"
events: [
  {
```

```
event_id: "evt_a1b2c3",
timestamp: "2025-01-15T10:30:00Z",
event_type: "page_view",
properties: { page: "/products", duration_ms: 4500 }
}
]
```

Response (202 Accepted):

accepted: 1, failed: 0, stream_id: "user-activity-prod"

3.2 Analytics Queries

POST /v2/analytics/query

Description: Execute analytical queries on ingested data

Rate Limit: 100 requests/minute

Timeout: 30 seconds

Request Body:

query: "SELECT event_type, COUNT(*) as count, AVG(properties.duration_ms) as avg_duration FROM events WHERE timestamp >= '2025-01-01' GROUP BY event_type ORDER BY count DESC LIMIT 20"

format: "json"

cache_ttl: 300

3.3 Real-time Aggregations

GET /v2/streams/{stream_id}/metrics

Parameters: window (1m|5m|1h|1d), metrics (count|sum|avg|p95|p99)

Description: Get real-time sliding window aggregations

Rate Limit: 1,000 requests/minute

4. ERROR HANDLING

Standard HTTP status codes with structured error bodies:

400 Bad Request:

error: { code: "INVALID_QUERY", message: "Syntax error at position 45", details: {...} }

401 Unauthorized:

error: { code: "AUTH_EXPIRED", message: "Token expired", action: "Refresh using /oauth/token" }

429 Too Many Requests:

error: { code: "RATE_LIMITED", message: "Rate limit exceeded" }

Headers: X-RateLimit-Remaining: 0, X-RateLimit-Reset: 1706234567, Retry-After: 42

500 Internal Server Error:

error: { code: "INTERNAL_ERROR", message: "...", request_id: "req_x7k2..." }

All 5xx errors include a request_id for support tickets. Our SLA guarantees 99.95% uptime.

5. WEBHOOKS

Configure webhooks to receive real-time notifications:

POST /v2/webhooks

Request Body:

```
url: "https://your-app.com/webhook"  
events: ["stream.anomaly_detected", "query.completed", "alert.triggered"]  
secret: "whsec_..."
```

Webhook payloads are signed using HMAC-SHA256. Verify signature:

Expected: HMAC-SHA256(webhook_secret, request_body)

Compare with: X-DataStream-Signature header

Delivery: At-least-once with exponential backoff (max 3 retries over 24h).

6. SDK SUPPORT

Official SDKs available:

Python: `pip install datastream-sdk (v2.4.0)`

Node.js: `npm install @datastream/sdk (v2.4.0)`

Go: `go get github.com/datastream/sdk-go (v2.4.0)`

Java: Maven artifact `com.datastream:sdk:2.4.0`

All SDKs include automatic retry, connection pooling, and built-in metrics.